Factor-base designs: A0′s memory-bound cells on the sparse tower engine (run-20261019T1632Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only: nothing here prices a pipeline, reports S or shows a speedup. **Protocol:** PREREGISTRATION.md Amendment 6 (registered in commit `02e91c9d4` before any T1 or T2 row), with Amendments 1–5 for everything it does not change.

Why this run

Two A0 cells are missing under `f4_fp` for memory: m = 3, t = 5 (F3; the cell where A11 finishes at D = 5) and m = 4, t = 4 (C2 at 13 GB, G1 at 15 GB). The repository's sparse tower engine `f4_fp_tower` needs far less memory. Amendment 6 admits it as a second instrument for A0 only if it first reproduces `f4_fp`'s D on every completed planted A0 system (T1), and only then measures the two missing cells (T2).

Run

- **Commands:** `fb_designs_slope admit-tower research/factor_base_designs_20261015/runs` (T1), then `fb_designs_slope run --engine tower --arms A0 --m 3 --t-min 5 --t-max 5 --budget 3600 --grace 300 --vmem-kb 15000000 --file 'T2-{arm}-m{m}'` and the same with `--m 4 --t-min 4 --t-max 4` (T2).
- **Commits measured:** T1 at `02e91c9d413cbaba6004748b40913ca46bdc91c7`, T2 at `0cc454e9c49d1d55dd789a6bec52c9ec50940bfb`, recorded in each row.
- **Host:** "vm", Intel Xeon @ 2.10 GHz, 4 cores, 15 GB, no swap, a Claude Code cloud container. One system at a time.
- **Window:** 2026-10-19, T1 16:30Z to 16:31Z; T2 16:32Z to 18:44Z (UTC).
- **Engine:** A0′s Kummer equations become the ring's square rules (`y_{j+1} = y_j²`, `y_{t−1}² = 1`). The summation equations, reduced into the ring, are the input. Seeds and constructions are A0′s own, so the systems are identical to the `f4_fp` ones. Every equation is checked at the planted solution, and every tower equation is checked to vanish in the ring, before the engine runs.
- **Records:** `runs/T1-admission.jsonl`, `runs/T2-A0-m3.jsonl`, `runs/T2-A0-m4.jsonl` and their `.log` files, and `runs/analysis-tower.txt` (native `analyze --tower`).

T1: admission

Admitted. All 26 completed planted A0 systems on p_1 (the first run's m = 2, t = 2–9 and m = 3, t = 2–4, and F4′s m = 4, t = 2, 3) complete on the tower engine with the committed `f4_fp` D, without deriving 1, and with the planted solution kept in the basis. The engine took 0.002 s to 13.9 s per system.

The engine's `max_cols` is smaller than `f4_fp`'s at every one of them (for example 16 461 against 21 619 at m = 4, t = 3), and it is not the same quantity: the engines work in different rings. Matrix widths are not compared across engines in this study.

T2: the two cells

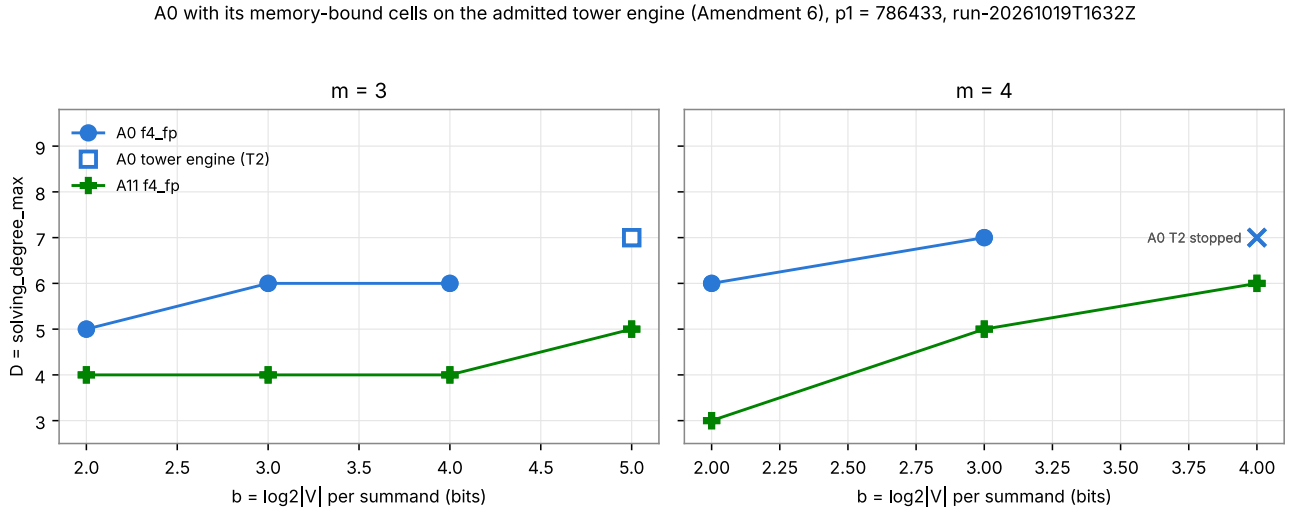


Figure 1: D against b for A0 at m = 3 (left) and m = 4 (right) on p_1: `f4_fp` cells, the T2 cells on the tower engine, and A11 (`f4_fp`) beside. Both targets per cell are plotted (n = 2; they coincide). The cross is the T2 m = 4 cell, at the partial D = 7 both systems reached. Run run-20261019T1632Z; rows from `runs/*.jsonl`, plotted by `plot.py`.

cell	system	D	degree reached	time (s)	tower-engine `max_cols`	status
m = 3, t = 5	target 0	7	8	383.5	34 706	ok
m = 3, t = 5	target 1	7	8	377.5	35 617	ok

m = 4, t = 4	target 0	7 (partial)	7	3600.1	88 925 (partial)	timeout at the 3600 s budget
m = 4, t = 4	target 1	7 (partial)	8	3600.4	170 036 (partial)	timeout at the 3600 s budget

- **m = 3, t = 5:** both systems complete, with D = 7. With this cell, A0′s m = 3 series reads D = 5, 6, 6, 7 on 4 completed cells: σ = 0.6000, **no middle ground** (§1). Without it (the 3 `f4_fp` cells), σ = 0.5000, the same verdict. A11 at the same cell is D = 5 (F3), so A11′s D is 2 below A0′s here (descriptive).
- **m = 4, t = 4:** both systems stopped at the 3600 s budget, at partial D = 7 (the second had reached degree 8). The memory limit did not bind: spot checks showed 6–8 GB resident against the 15 GB limit, where `f4_fp` aborted for memory after about 405 s. A cell with a system that did not complete does not join the fit (Amendment 6), so A0′s m = 4 series stays at 2 completed cells: σ = 1.0000, **inconclusive**, and its registered prediction "σ ≥ 0.5 at m = 4" remains **not decided**.

Predictions scorecard (Amendment 6)

prediction	observed	held?
T1: admitted	26 of 26 match	yes
T2, m = 3: completes with D = 7, so the series reads 5, 6, 6, 7 (σ = 0.6)	D = 7 on both systems; σ = 0.6000 on 4 cells	yes
T2, m = 4: does not complete within the limits	both systems stop at 3600 s, partial D = 7	yes

Limits of what this shows

- T1 checks agreement up to the sizes `f4_fp` could complete (m = 3, t ≤ 4; m = 4, t ≤ 3). That the engine's D at m = 3, t = 5 equals `f4_fp`'s is an assumption those 26 agreements support, not a measurement: `f4_fp` has no completed system there to compare.
- The tower engine's own cost counter (`muladds`) is recorded but is not `field_ops`, so κ (Amendment 4) is not extended by these cells.
- Times are wall-clock on one 4-core container. `f4_fp` and the tower engine ran under different limits here (memory against time), so no timing comparison is made between them.
- One cell and one prime. Nothing here replicates the m = 3, t = 5 reading on p_2.

Deviations

None from Amendment 6.